Event-Density Channels and the Emergence of Global Periods in High-Precision Scattering

ED Interpretation of: Calabi–Yau Manifolds in Black-Hole Encounters

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Abstract

High-precision calculations of black-hole scattering at fifth post-Minkowskian order reveal a striking result: the radiated energy and recoil depend on Calabi–Yau three-fold periods, global geometric invariants normally associated with string theory and algebraic geometry. In standard GR and QFT, these structures appear only in intermediate Feynman integrals and cancel out of physical observables. Their survival in the final radiative sector signals a deeper architectural constraint.

In the ED ontology, this constraint is natural. A close-pass black-hole encounter is a high-ED structural transition in which steep ED-gradients collapse the configuration space into a restricted ED-channel. Temporal tension accumulates during approach and is released globally as gravitational radiation. Because the system must evolve along the entire ED-channel, the observables inherit the global geometry of that channel. Calabi–Yau periods are the mathematical shadow of this geometry.

The conservative sector cancels CY contributions because it samples only local ED-structure. The radiative sector retains them because it reflects global ED-relaxation. ED therefore predicts not only the appearance of CY3 periods at 5PM, but also a hierarchy of higher-dimensional periods at stronger ED regimes.

This paper unifies the gravitational result with ED-I-013's reinterpretation of quantum information. In Deutsch-style oracles, the system is forced into a high-multiplicity ED-channel whose global structure encodes a period extracted in one step. In black-hole scattering, the system is forced into a high-ED structural channel whose global structure encodes a Calabi–Yau period revealed in the radiated energy. Both phenomena arise from the same architectural principle: global invariants emerge whenever ED-gradients restrict the configuration space.

1. Introduction: High-ED Transitions Reveal Hidden Global Geometry

Black-hole scattering is one of the most extreme structural events the universe produces. When two compact objects pass near one another at relativistic speeds, the local event density spikes, temporal tension builds and releases, and the ED-gradients around the encounter become sharply curved. These transitions are not incremental; they are global reorganizations of ED-structure. In such regimes, the system is no longer free to explore arbitrary configurations. It is forced into a narrow structural channel whose geometry determines every observable outcome.

The recent work of Driesse, Jakobsen, Klemm, Mogull, Nega, Plefka, Sauer, and Usovitsch demonstrates this with unprecedented clarity. Their 5PM–1SF calculation of black-hole scattering reveals that the radiated energy and recoil depend on Calabi–Yau three-fold periods—global geometric invariants of a highly constrained manifold. These periods are not decorative mathematical artifacts. They are the signature of a system whose allowable histories lie on a restricted geometric structure.

In the ED ontology, this is exactly what high-ED transitions should produce. When ED-gradients become extreme, the configuration space collapses into a structural channel. The global geometry of that channel determines the

integrals that appear in the calculation. The emergence of Calabi–Yau periods is therefore not surprising. It is the mathematical shadow of the ED-channel that governs the scattering event.

This paper develops the ED interpretation of the Driesse et al. result. It shows how Calabi–Yau periods arise naturally from ED-channels, how temporal tension and ED-gradients shape the radiative sector, and why global invariants appear only when the system undergoes a high-ED reconfiguration. It also connects this phenomenon to ED-I-013, where the same architectural mechanism explains the global periodicity extracted by Deutsch-style quantum oracles. In both domains—black-hole scattering and quantum information—the system is forced into a constrained manifold, and the global invariants of that manifold appear directly in the observables.

(Parenthetical geometric cue: locally, the ED-channel resembles a saddle—one direction compresses as the objects approach, another expands as radiation carries structure away. Globally, this channel wraps into a manifold whose periods encode the allowed histories.)

2. ED Ontology: The Structural Components Relevant to Scattering

The ED interpretation of high-precision scattering requires only the core elements of the ontology—event density, temporal tension, ED-gradients, and channel geometry. These components determine how a system evolves when its internal and external ED-structure becomes extreme.

Event Density and ED-Gradients

Every physical system is embedded in an ED-field whose gradients determine what transitions are possible. In low-ED regimes, these gradients are shallow and the system can explore many configurations. In high-ED regimes, the gradients become steep, sharply constraining the system's allowable histories. Black-hole encounters are among the highest-ED events in the universe, and their gradients are correspondingly restrictive.

Temporal Tension

Temporal tension is the ED-based quantity that replaces dark matter in galactic dynamics and governs the release of energy in high-ED transitions. In scattering, temporal tension accumulates as the objects approach and is released as gravitational radiation. Radiated energy and recoil are therefore direct expressions of how temporal tension flows through the ED-channel.

ED-Channels as Restricted Configuration Manifolds

An ED-channel is the structural pathway through which a system evolves when ED-gradients restrict its degrees of freedom. In high-ED events, the channel becomes narrow and highly structured. The system cannot wander; it must follow the channel. The global geometry of this channel determines the integrals that appear in the calculation.

This is the ED mechanism behind the emergence of Calabi–Yau periods. When the channel's global geometry matches the structure of a Calabi–Yau three-fold, the periods of that manifold appear in the radiated energy and recoil.

Global Geometry and Structural Invariants

ED-channels encode global constraints. When a system is forced to follow the entire channel, the global invariants of that channel become observable. In quantum information, this produces the global period extracted by Deutsch-style oracles. In black-hole scattering, it produces Calabi–Yau periods. In both cases, the invariant is not a dynamical accident but a structural necessity.

3. Black-Hole Encounters as High-ED Structural Transitions

A close-pass encounter between two compact objects is one of the highest-ED events the universe produces. As the black holes approach, the local event density rises sharply. Temporal tension accumulates, ED-gradients steepen, and the system enters a regime where its allowable transitions are no longer flexible. The encounter becomes a structural transition, not a dynamical trajectory.

In ED, a black-hole scattering event is not two masses moving through spacetime. It is the reconfiguration of ED-structure under extreme gradients. The approach phase compresses ED-structure, the near-pass forces a global reorganization, and the departure phase releases temporal tension as gravitational radiation. Every observable—scattering angle, radiated energy, recoil—is a measurement of how the ED-structure deforms and relaxes.

The key point is that high-ED transitions collapse the configuration space. The system cannot explore arbitrary histories. It is forced into a narrow structural channel whose geometry determines the evolution. This is why the scattering problem becomes mathematically rigid at high precision: the ED-channel is rigid.

The Driesse et al. result reveals this rigidity directly. The appearance of Calabi–Yau periods in the radiative sector is the mathematical imprint of a system whose ED-channel has a nontrivial global geometry. The scattering event is not sampling a large configuration space; it is following a single, highly constrained manifold.

This is the ED interpretation:
Black-hole scattering is a high-ED structural transition confined to a global channel whose geometry determines the integrals that appear in the calculation.

4. Why Restricted Configuration Manifolds Produce Global Invariants

When ED-gradients become extreme, the system's configuration space collapses. The ED-field prunes away all but a narrow set of allowable transitions. The system is forced to evolve along a restricted manifold—an ED-channel. This restriction is not dynamical; it is architectural. It arises because the ED-structure cannot support arbitrary histories under extreme gradients.

A restricted manifold has global invariants. These invariants are not optional. They are built into the topology and geometry of the channel. When the system is forced to follow the entire channel, those invariants appear directly in the observables.

This is why the Calabi–Yau periods appear in the radiated energy and recoil. The radiative sector is the part of the scattering event where the ED-structure undergoes a global reconfiguration. Temporal tension is released along the entire channel, and the integrals describing that release inherit the global geometry of the channel. The Calabi–Yau periods are the global invariants of that geometry.

The conservative sector, by contrast, does not involve global reconfiguration. It is dominated by local ED-structure, and its contributions cancel the Calabi–Yau terms. This is exactly what ED predicts:

- Local behavior → polylogarithms, elliptic integrals, cancellations.
- Global behavior → Calabi–Yau periods, higher-dimensional invariants.

The ED principle is simple:

When ED-gradients restrict the configuration space, the system's observables inherit the global invariants of the ED-channel.

This is the same architectural mechanism that appears in ED-I-013. In the Deutsch oracle, the system is forced into a high-multiplicity ED-channel whose global structure encodes a period. The algorithm extracts that period in one step because the channel already encodes it. The scattering problem is the gravitational analogue: the ED-channel encodes a Calabi–Yau period, and the radiated energy reveals it.

The architecture is the same.
The domain is different.
The invariants are universal.

5. ED Interpretation of Calabi–Yau Periods in the 5PM Radiation Sector

The central surprise of the Driesse et al. result is that Calabi–Yau three-fold periods appear directly in the radiated energy and recoil of a black-hole scattering event. In standard GR or QFT language, this is unexpected: physical observables are usually simpler than the intermediate integrals. Yet here, the global geometry of a Calabi–Yau manifold survives all cancellations and enters the final answer.

In ED, this is exactly what should happen.

A black-hole encounter is a high-ED transition. Temporal tension accumulates as the objects approach, and the ED-gradients become steep enough to collapse the configuration space into a restricted ED-channel. When the system radiates, it is not releasing energy arbitrarily—it is releasing temporal tension along the entire global channel. The integrals describing this release must therefore inherit the global geometry of the channel itself.

Calabi–Yau periods are the mathematical expression of that geometry.

They appear because:

- the ED-channel has a nontrivial global structure,
- the radiative sector samples the full channel,
- and the integrals over that channel reduce to periods of a CY3 manifold.

This is not a coincidence. It is the architectural signature of a system whose allowable histories lie on a constrained manifold.

The conservative sector does not show CY periods because it does not involve global reconfiguration. It is dominated by local ED-structure, and its contributions cancel. Only the radiative sector—where temporal tension flows across the entire channel—reveals the global invariant.

In ED terms:

- Calabi–Yau periods appear when the ED-channel is global, not local.
- Radiation is global. The conservative sector is not.

This is why the CY3 periods appear exactly where they do, and nowhere else.

6. Temporal Tension, ED Gradients, and the Origin of Radiated Energy

Radiated energy and recoil are not mysterious byproducts of gravitational scattering. In ED, they are direct expressions of how temporal tension flows during a high-ED transition.

Radiated Energy as Temporal Tension Release

As two compact objects approach, the ED-gradients steepen and temporal tension accumulates. At closest approach, the ED-structure is maximally compressed. When the objects separate, the ED-structure relaxes, releasing temporal tension as gravitational radiation.

This release is not local. It is a global relaxation along the entire ED-channel. The integrals describing this relaxation must therefore reflect the channel's global geometry. This is why the radiated energy depends on Calabi–Yau periods.

Recoil as Asymmetry in ED-Gradients

Recoil arises when the ED-gradients are not symmetric across the channel. If more temporal tension is released in one direction than another, the remnant black hole receives a kick. The same CY periods appear in the recoil because the asymmetry is encoded in the same global channel.

In ED:

- Energy release = global relaxation of temporal tension.
- Recoil = directional imbalance in that relaxation.
- Both depend on the same global geometry.

Why the Conservative Sector Cancels CY Contributions

The conservative sector does not involve global ED-reconfiguration. It is dominated by local gradients and local curvature. Local behavior produces polylogarithms, elliptic integrals, and cancellations. Only when the system undergoes a global transition—the radiative phase—do the global invariants survive.

ED Summary of the Radiative Sector

High-ED encounter → steep gradients → restricted channel.
Restricted channel → global geometry → Calabi–Yau manifold.
Temporal tension release → integrals over the full channel → CY periods.
Recoil → asymmetry in the same channel → same CY periods.

In ED, the appearance of Calabi–Yau periods is not surprising. It is inevitable.

7. The ED Prediction: When Global Periods Should Appear

High-precision scattering exposes a hierarchy of mathematical structures—polylogarithms, elliptic integrals, K3 periods, and now Calabi–Yau three-fold periods. In standard GR and QFT, this hierarchy is treated as an emergent complexity of Feynman integrals. In ED, the hierarchy is structural. It reflects the degree of ED-restriction imposed by the event.

The ED principle is simple:

The stronger the ED-gradients, the narrower the ED-channel, and the higher the dimensionality of the global invariant that appears in the observables.

This yields a clean, predictive ladder:

- Low-ED transitions
  - Shallow gradients → broad configuration space → polylogarithms.
  - (Local ED-structure only; no global channel.)
- Moderate-ED transitions
  - Stronger gradients → restricted channel → elliptic periods.
  - (The channel has one nontrivial cycle.)
- High-ED transitions
  - Very steep gradients → narrow channel → K3 periods.
  - (Two-cycle global geometry.)
- Extreme-ED transitions
  - Maximal gradients → fully constrained channel → CY3 periods.
  - (Three-cycle global geometry.)
- Ultra-extreme transitions
  - ED predicts higher-dimensional periods (CY4, CY5…) in regimes beyond 5PM.
  - (The channel becomes a higher-dimensional manifold.)

This is not a mathematical coincidence. It is the structural signature of how ED-channels behave under increasing tension.

(Parenthetical cue: the saddle is the local shape of the channel; the Calabi–Yau period is the global signature of the channel.)

The Driesse et al. result fits perfectly into this hierarchy. The radiative sector of a 5PM scattering event is an extreme-ED transition. The ED-channel is fully global, and the integrals inherit the geometry of a CY3 manifold. ED therefore predicts:

- CY3 periods should appear in any observable tied to global ED-relaxation.
- Higher-dimensional CY periods should appear at higher PM orders or stronger mass-ratio regimes.
- Bound-state analytic continuation should preserve the same global invariants.

The appearance of Calabi–Yau periods is not surprising. It is the expected outcome of an ED-channel with global structure.

8. Connection to ED-I-013: Global Periodicity in Deutsch-Style Oracles

The emergence of Calabi–Yau periods in black-hole scattering is not an isolated phenomenon. It is the gravitational analogue of the global periodicity extracted in Deutsch-style quantum algorithms. ED-I-013 showed that:

- A high-multiplicity ED-channel supports global participation.

- The oracle applies a single participation rule to the entire channel.
- The global structure of the channel encodes a period.
- The algorithm extracts that period in one step because the channel already contains it.

The same architecture appears in black-hole scattering.

The shared ED mechanism
Both systems—quantum oracles and black-hole encounters—are governed by the same ED principle:

When ED-gradients restrict the configuration space, the system is forced into a global channel whose invariants appear directly in the observables.

- In quantum information:
- The channel is a high-multiplicity participation structure.
- The invariant is a period of the function's global geometry.

The QFT layer resolves the symmetry of the channel.

- In black-hole scattering:
- The channel is a high-ED structural manifold.
- The invariant is a Calabi–Yau period of the channel's global geometry.

The radiative sector resolves the symmetry of the channel.

Why the analogy is exact
Deutsch's oracle does not "evaluate all inputs simultaneously."
It applies one participation rule to a global ED-channel.

The scattering event does not "explore all Feynman histories."
It evolves along a global ED-channel.

The period in Deutsch's algorithm is not a quantum trick.
It is a global invariant of the channel.

The Calabi–Yau period in scattering is not a mathematical accident.
It is a global invariant of the channel.

The architectural unification
Quantum information and gravitational scattering are not separate domains. They are two manifestations of the same ED architecture:

- Global access
- Global constraint extraction
- Global symmetry resolution
- Global invariants encoded in ED-channels

The appearance of Calabi–Yau periods in scattering is the gravitational counterpart of the period-extraction phenomenon in quantum algorithms. Both arise because ED-channels encode global structure.

9. Architectural Unification Across Domains

The emergence of Calabi–Yau periods in black-hole scattering and the emergence of global periods in Deutsch-style quantum algorithms are not parallel curiosities. They are two expressions of the same architectural principle: when ED-gradients restrict the configuration space, the system is forced into a global channel whose invariants appear directly in the observables.

In quantum information, ED-I-013 showed that:

- high-multiplicity channels support global participation,
- the oracle applies a single participation rule to the entire channel,
- the channel's global structure encodes a period,
- and the QFT layer resolves that global symmetry in one step.

In black-hole scattering:

- high-ED gradients collapse the configuration space into a restricted structural manifold,
- the radiative sector forces the system to traverse the entire global channel,
- and the integrals describing temporal-tension release reduce to Calabi–Yau periods.

These are not different mechanisms. They are the same mechanism instantiated in different physical regimes.

(Parenthetical cue: the saddle is the local shape of the channel; the period—computational or geometric—is the global signature of the channel.)

The architectural unification is straightforward:

- Global access (Deutsch) ↔ Global ED-reconfiguration (scattering)
- Global constraint extraction ↔ Global channel restriction
- Symmetry extraction ↔ Period extraction
- Participation geometry ↔ Calabi–Yau geometry

The mathematics differs because the channels differ.
The architecture is identical because ED-channels obey the same structural rules across domains.

This unification is not metaphorical. It is predictive.
If ED-channels govern both quantum information and gravitational scattering, then global invariants—periods, symmetries, cycles—should appear wherever ED-gradients restrict the configuration space.

The Driesse et al. result confirms this prediction in the gravitational domain.
ED-I-013 confirms it in the computational domain.
Together, they reveal a single architectural principle underlying two fields that are normally treated as unrelated.

10. Predictions, Tests, and Future Directions

The ED interpretation of Calabi–Yau periods in black-hole scattering is not descriptive; it is predictive. If the appearance of CY3 periods is the signature of a global ED-channel, then ED makes several concrete predictions about where such structures should appear next.

Higher-Order PM Predictions
As the PM order increases, ED-gradients become more restrictive and the ED-channel becomes more global. ED therefore predicts:

CY4 and CY5 periods should appear at higher PM orders, especially in the radiative sector.

The conservative sector will continue to cancel global invariants unless the ED-channel itself becomes global in that regime.

Mass-Ratio Predictions
The Driesse et al. result includes only the subleading mass-ratio term. ED predicts:

- Comparable-mass systems will show stronger CY signatures because the ED-channel becomes more symmetric and more global.
- Next-to-next-to-leading mass-ratio terms should reveal new Calabi–Yau families, as the authors themselves anticipate.

Bound-State Predictions
Analytic continuation from scattering to bound orbits should preserve the global invariants of the ED-channel. ED predicts:

- CY3 periods should appear in eccentric inspiral waveforms, especially in the tail-of-tail sector.
- The transition from scattering to inspiral should not erase the global geometry.

Astrophysical Predictions
Any high-ED astrophysical transition should reveal global invariants:

- extreme tidal encounters,
- near-horizon scattering,
- high-velocity flybys in dense clusters,
- and possibly early-universe transitions.

Wherever the ED-channel becomes global, its periods should appear.

Quantum Information Predictions
Because ED-channels unify gravitational and computational phenomena:

- new quantum algorithms should emerge from global symmetry extraction,
- new cryptographic primitives should arise from channel incompatibility,
- and new error-correcting codes should follow from distributed alignment.

These predictions extend ED-I-013 into the gravitational domain and extend the Driesse et al. result into the computational domain.

The Architectural Claim

The appearance of Calabi–Yau periods in black-hole scattering is not an isolated mathematical curiosity. It is the first direct evidence that ED-channels possess global geometry, and that this geometry governs both gravitational and computational phenomena.

The next step is to test these predictions in higher-order PM calculations, in comparable-mass regimes, and in eccentric inspirals. If the same global invariants continue to appear, the ED interpretation will not merely explain the Driesse et al. result—it will generalize it.

References:

Driesse, M., Jakobsen, G. U., Klemm, A., Mogull, G., Nega, C., Plefka, J., Sauer, B., & Usovitsch, J. (2025). *Emergence of Calabi–Yau manifolds in high-precision black-hole scattering*. Nature, 641(8063), 603–610